

CONNECTED VEHICLE TELEMATICS & RISK ANALYTICS

Real-Time Sensor Diagnostics, Predictive Quality & Dynamic Insurance Pricing

CIA 3 Project Report | Jaidev Prasad (25121019) • Ruth Wilson (25121034) • Bala Kiran (25121012)

EXECUTIVE PROJECT SUMMARY

This project demonstrates an end-to-end automobile telematics framework that captures connected-car sensor data (braking deceleration, speeding, late-night driving, engine coolant temp, oil pressure, and brake wear) to score individual driver safety, predict component failures, eliminate fraud, and generate fair dynamic premiums.

FLEET SIZE 15 Vehicles 4G Connected	AVG SCORE 57.3 / 100 Pricing Driver	CLAIM PROB 20.9% Fleet Baseline	RENEWAL HIKE 46.3% Trajectory	AVG PREMIUM Rs. 15,933 vs Rs. 18,000	TOTAL SAVINGS Rs. 31,000 Safe Drivers
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1. MATHEMATICAL SCORING & DYNAMIC PRICING ENGINE

A. Driving Safety Score (0 - 100 points):

Score = 100 - (Harsh_Brakes * 3) - (Speeding_Violations * 4) - (Night_Driving_Hours * 2)

B. Dynamic Premium Calculation Formula:

Dynamic Premium = Base_Rate (Rs. 15,000) + Score_Risk_Adjustment + Maintenance_Alert_Fee (Rs. 1,000)

Driving Score Tier	Risk Profile	Base Adjustment	Dynamic Premium	Claim Prob.	Hike Chance
Score >= 80	Safe Driver (Low Risk)	-Rs. 3,000 Discount	Rs. 12,000 / yr	3.8% - 6.2%	< 5% (Locked Discount)
Score 60 - 79	Average Driver (Moderate)	Rs. 0 (Standard Base)	Rs. 15,000 / yr	10.5% - 18.0%	25% - 45%
Score < 60	Risky Driver (High Risk)	+Rs. 3,000 Surcharge	Rs. 18,000 - 19,000	38.5% - 48.0%	88% - 98% (High Surcharge)

2. ENGINE & COMPONENT QUALITY TELEMETRY MATRIX

1. Cooling System (ECT):	Normal: 85C - 95C Alert: > 100C Alert	Use: Prevents blown head gaskets, engine seizure & false warranty claims.
2. Combustion & Oil PSI:	Normal: 30 - 60 PSI Alert: < 20 PSI Alert	Use: Detects low lubrication, aggressive RPM redlines & premature bearing wear.
3. 12V Battery Health:	Normal: 12.4V - 12.8V Alert: < 12.1V Alert	Use: Predicts alternator failure, deep discharge & roadside dead battery calls.
4. Braking Wear Sensor:	Normal: > 5.0 mm Alert: < 3.0 mm Alert	Use: Correlates high-G deceleration stops (-0.74g) with brake pad degradation.
5. TPMS Tire Pressure:	Normal: 32 - 35 PSI Alert: < 28 PSI Alert	Use: Eliminates high-speed highway tire blowouts and reduces rolling friction.

3. AUTOMOTIVE INDUSTRY STRATEGIC FRAMEWORK (CIA-3 ALIGNMENT)

Pay-How-You-Drive (PHYD) Isolates real-time driver behavior from static demographics, mirroring Zuno SmartDrive & Tesla Safety Score models. IRDAI regulatory sandbox compliance.	Predictive Quality Savings Connects to Slide 7 (We Predict £46B global warranty claims benchmark and Mahindra & Mahindra SAS Field Quality Weibull reliability analytics).
Dual Risk & LGD Reduction Collateral tracking lifts recovery from 60% to 70%, reducing Loss Given Default (LGD) by 25% without altering borrower credit score (Slide 2).	Fraud Event Forensics Uses deceleration G-forces and timestamps to verify genuine mechanical failure vs staged fraud rings (e.g. Insure The Box telematics forensics).

CONNECTED VEHICLE TELEMETRY & SENSOR MASTER DATASET

Complete 15-Vehicle Fleet Ledger with Engine Diagnostics, Risk Scores & Renewal Hike Probabilities

4. COMPLETE 15-VEHICLE TELEMATICS LEDGER

ID	Driver	Car Model	Coolant	Oil PSI	RPM	Pad	Score	Claim %	Hike %	Premium	Savings
CAR-01	Aarav Sharm	Hyundai Creta	88C	45	3200	9.2mm	84	6.2%	4.0%	Rs. 12,000	+Rs. 6,000
CAR-02	Priya Nair	Maruti Brezza	102C	28	5100	2.8mm	32	38.5%	88.0%	Rs. 19,000	-Rs. 1,000
CAR-03	Rohan Mehta	Tata Nexon EV	84C	50	2800	10.5mm	93	3.8%	2.0%	Rs. 12,000	+Rs. 6,000
CAR-04	Sneha Kulka	Mahindra XUV70	105C	22	5400	2.2mm	20	44.0%	94.0%	Rs. 19,000	-Rs. 1,000
CAR-05	Vikram Sing	Honda City ZX	89C	42	3400	6.8mm	70	12.5%	25.0%	Rs. 15,000	+Rs. 3,000
CAR-06	Ananya Roy	Kia Seltos GTX	92C	38	3900	5.4mm	61	16.0%	35.0%	Rs. 15,000	+Rs. 3,000
CAR-07	Karan Patel	Tata Punch Cre	87C	44	3100	8.8mm	86	5.5%	5.0%	Rs. 12,000	+Rs. 6,000
CAR-08	Neha Gupta	Maruti Baleno	100C	25	4900	3.0mm	21	42.0%	92.0%	Rs. 19,000	-Rs. 1,000
CAR-09	Aditya Verm	Toyota Urban C	88C	46	3300	7.5mm	75	10.5%	18.0%	Rs. 15,000	+Rs. 3,000
CAR-10	Divya Menon	Hyundai Venue	94C	36	4100	4.8mm	50	22.0%	60.0%	Rs. 18,000	+Rs. 0
CAR-11	Siddharth D	Mahindra Thar	106C	18	5600	2.0mm	20	48.0%	98.0%	Rs. 19,000	-Rs. 1,000
CAR-12	Pooja Reddy	Skoda Kushaq S	86C	48	3000	9.5mm	89	4.5%	3.0%	Rs. 12,000	+Rs. 6,000
CAR-13	Rahul Deshm	Volkswagen Tai	91C	40	3800	5.8mm	59	18.0%	45.0%	Rs. 18,000	+Rs. 0
CAR-14	Tanvi Joshi	Tata Harrier F	103C	24	5000	2.9mm	20	43.0%	90.0%	Rs. 19,000	-Rs. 1,000
CAR-15	Manish Chop	Honda Elevate	88C	44	3200	8.5mm	79	8.0%	10.0%	Rs. 15,000	+Rs. 3,000
FLEET AVERAGE			92.8C	36.9	4180	5.7mm	57.3	20.9%	46.3%	Rs. 15,933	Tot: Rs. 31k

CAR-02 (Priya Nair - Maruti Brezza) • 04-Sep 23:45:12 Decel G-Force: -0.68g ABS Active Stop Speed Drop: 88 -> 12 km/h at Bandra Sea Link Exit Telemetry Assessment: Tailgating emergency stop. Brake pad thickness eroded to 2.8mm, triggering component alert fee.
CAR-04 (Sneha Kulkarni - Mahindra XUV700) • 07-Sep 00:15:22 Decel G-Force: -0.72g Automatic FCW Brake Speed Drop: 105 -> 20 km/h at E-City Elevated Toll Telemetry Assessment: Severe collision warning trigger. Engine coolant hit 105C and oil pressure dropped to 22 PSI at 5,400 RPM.
CAR-11 (Siddharth Das - Mahindra Thar) • 03-Sep 01:30:15 Decel G-Force: -0.74g High G-Decel Speed Drop: 110 -> 25 km/h at Bor Ghat Descent Telemetry Assessment: High-speed night run (134 km/h peak). DTC P0217 Overheating logged; 98% chance of renewal hike.

6. PROJECT SUBMISSION DELIVERABLES

1. Native Excel Workbook:	Simple_Car_Sensor_Insurance_Analysis.xlsx (Multi-sheet, formulas =100-(E*3)-(F*4)-(G*2), AVERAGE, SUM)
2. Interactive HTML Dashboard:	index.html (10 interactive charts, live driver simulator, vehicle profile drawer, real-time probability meters)
3. Telemetry CSV Dataset:	detailed_telematics_sensor_data.csv (Complete raw sensor data with VIN, route, speed, temp, and probabilities)
4. Project Presentation Alignment	CIA 3 Automobile Industry: Jaidev Prasad (25121019), Ruth Wilson (25121034), Bala Kiran (25121012)
5. Live Local Server URL:	http://localhost:8000 (Open in any web browser to view interactive visual intelligence gallery)